

Ultrafast X-Ray Experiments

**Virtual Experiments at the X-Ray Laser European XFEL
Versuch EXP21**

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1. Einleitung

2. Theoretical Background

2.1. Synchrotronradiation

When accelerating electrons or other charged particles to relativistic speeds along a curved trajectory they emit so-called synchrotron radiation. It gains its name from the ring accelerators, the synchrotrons, where it was first observed. A typical synchrotron facility consists of a linear accelerator leading into a booster ring in which the electrons are brought up to relativistic speeds [1] [3]. The electrons are then fed into a storage ring in which they continue to circulate at relativistic speeds. From the storage ring the beam-lines lead tangentially away in which the synchrotron radiation is utilized in experimental hutches [1] [3]. Synchrotron radiation is bright X-radiation with a wide tunability in energy, which is useful for examining structures and dynamics of materials, as x-ray photons have energies in the order of several kiloelectronvolts (keV) [3]. This means their wavelengths are in the order of magnitude of Angstroms and comparable to the spacing of atoms in solid materials. This is useful, as it means that atoms in a crystalline array can diffract x-ray beams and the interference patterns can be measured [1].

There are four generations of synchrotrons with the main difference being the brilliance achieved [1]. The first was the bending magnet in the synchrotron as it was used for high energy collider physics with no additional modifications. The second generation introduced the wiggler, a configuration of dipole magnets with alternating polarity to achieve synchrotron radiation on a linear path. The third generation synchrotron radiation facilities were designed for low emittance of the beams with more effective linear magnet configurations, the undulators [3]. A low emittance means a smaller angular divergence of the beam and a higher brilliance [3]. The fourth generation in synchrotron radiation are the so-called free electron lasers (FELs)

2.1.1. Free Electron Laser (FEL)

2.1.2. Undulators

2.2. The investigated Molecule

2.2.1. Electronic Structure of $[Fe(bipy)_3]^{2+}$

2.2.2. Photophysics

2.3. Pump-Probe-Principle

2.3.1. The Pump-Pulse

2.3.2. The Probe-Pulse

3. Experimental Setup

3.1. Software: VLab

Datenstruktur (beam imaging and xray eye): 420x420 pixel, bei beam screen dim von 10mm x10mm, bei xray eye screen 2mm x 2mm

3.2. Undulator and Beamline

3.3. Monochromator and Focusing Optics

monochromator focusing optics: mirrors, power slits, solid attenuators, refractive Lenses, X-ray eye

3.4. Spectrum Analyser

diffraction gratings

3.5. Time-arrival Detector

3.6. Sample Environment

3.7. Laser System

3.8. Von Hamos Spectrometer

Quelle Versuchsmappe:[1]

4. Experimental Procedure

4.1. Preparation of the Experiment

4.1.1. Alignment and Characterization of the X-ray Beam

Before doing anything on a beam line it's important to characterize it.

As the absorption edge for the Fe $K\beta_{1,3}$ line of 7065 eV, the energy of the xray beam is set to 7400 eV, to avoid the ripples near the reaction (siehe ??) but without too strong of a loss in intensity. To measure the initial X-ray beam the power slits are fully opened and the following attenuator foils are used to bring the beam to an intensity just below saturatuion of the beam imaging unit 2 (BIU2):

The size and position of the x-ray is now measured using BIU2. Now the four-bounce crystal-monochromator is turned on and then off again to record the reduction of the spectral bandwidth. Then the dispersive single-shot spectrum analyser is used to record the spectrum of the pink X-ray beam with a bragg angle of 24° .

For the following experiment the monochromatic beam is used as it leads to a higher resolution than the pink x-ray, even though the intensity of the beam suffers.

The last step of diagnosing the beam line is using the refractive beryllium x-ray lenses to focus the beam at the imaging unit x-ray eye. It's important to appropriately adjust the attentuator foils as to not burn the imaging unit with a focussed beam.

With beryllium lenses 1, 2 and 5 (taken from ??) and the attentuator foils listed here ??, a beam with the FWHM of 13 μm (siehe ??) is achieved. The total focus length is calculated with $\delta \approx 7 \cdot 10^{-6}$ [?] and the properties of the lenses themselves taken from ??:

$$F = \frac{R}{2\delta N} \Leftrightarrow \frac{1}{F_{ges}} = \frac{2\delta N_1}{R_1} + \frac{2\delta N_2}{R_2} + \frac{2\delta N_5}{R_5}$$
$$\Leftrightarrow F_{ges} = \frac{1}{2\delta(\frac{N_1}{R_1} + \frac{N_2}{R_2} + \frac{N_5}{R_5})} \approx 4,76 \text{ m}$$

—Notes—

X-ray energy $E_0 = 7400 \text{ eV}$ (has to be a good bit away from reaction edge of 7065 keV to be stable)

Alignment: attenuators to not burn through out screens (here: 4: B4C 0.8, 3: C 0.4, 2: Si 0.05, C 0.1)
size and position of the beam plots (FWHM) (jitter neu!) pink vs mono beam 2D/3D plots

optical resolution: 25 $\mu\text{m}/\text{px}$

Characterization: spectrum of the pink vs mono beam plots myb noch ex 7+8, aber idk

Focusing: plots of the focused beam size and position (FWHM)

4.1.2. Preparatory Estimates for the Experiment

Fun Facts: molar mass: $M([Fe(bipy)_3]^{2+}) \approx M([Fe(bipy)_3]) = 524,409 \frac{\text{g}}{\text{mol}}$ concentration of the sample: $c = 0,380 \frac{\text{mol}}{\text{L}}$ total attenuation of X-rays: attenuation comparison plot flow speed of the jet: $v_{jet} = 11 \frac{\text{m}}{\text{s}}$

Laser: excitation wavelength: $\lambda_{pump} = 400 \text{ nm}$ (dazu alle fun facts mit MLCT Zuständen und so, und nicht overshooten lol) focussing, diameter of the focus Overfilling: beam sizes of the optical laser and the X-ray pulse relative to each other (plot beam diameter)

Spectrometer: Van Hamos Spectrometer vs Johann Spectrometer (plot of the two geometries) crystal selection: $Si(531)$ vs $Ge(531)$ (plot of the two crystal reflections)

4.2. Performing the Experiment with Data Acquisition

5. Analysis and Interpretation

6. Discussion of the physical meaning

7. Conclusion and Outlook

References

- [1] P. Willmott, *An Introduction to Synchrotron Radiation*. Wiley, 2019.

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Appendix B Code